

# How to Create New Pickleball Lesson Handouts

This guide explains how to use the Pickleball Lesson Handouts project to create consistent student-facing PDF handouts.

## Project Location

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GitHub repository:

<https://github.com/londonj/PickleballLessonHandouts>

Local folder on the computer where this was set up:

```
C:\Coding\Pickleball_Lesson_Handouts
```

## What This Project Does

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This project creates polished PDF handouts for pickleball lessons using a consistent format.

It keeps the content and design separate:

- The lesson content is written in Markdown.
- The visual style is controlled by one shared CSS file.
- A Python script converts the Markdown into a PDF.
- Preview images are generated so the final layout can be visually checked before sharing.

This helps every handout use the same fonts, colors, section styling, spacing, and page-break rules.

## Important Files and Folders

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```
templates/student-handout-template.md
```

This is the full instructional template. It explains what each section should contain.

Use this when you need guidance on how to write the handout.

```
templates/new-handout-starter.md
```

This is a blank starter handout with all required section headings already included.

Use this when creating a new handout from scratch.

```
templates/handout-style.css
```

This controls the PDF design:

- fonts
- font sizes

- colors
- section heading bars
- margins
- spacing
- link color
- page-break behavior

Do not manually style individual PDFs. If the visual design needs to change, update this file so all future handouts stay consistent.

#### `scripts/build-handout-pdf.py`

This script converts a Markdown handout into a styled PDF.

It also creates preview images in the `work` folder so the PDF can be checked visually.

#### `prompts/create-student-handout.md`

This contains the prompt to use with an AI assistant when generating a new handout.

#### `docs/production-checklist.md`

Use this checklist before sharing a PDF with students.

#### `examples/`

Completed Markdown handouts can be stored here.

There is already an example:

#### `examples/third-shot-drop-student-handout.md`

#### `outputs/`

Finished PDF files are saved here.

#### `work/`

Generated HTML and preview PNG files are saved here.

This folder is for review and troubleshooting. It is ignored by Git and does not need to be shared.

## Step-by-Step Workflow

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### Step 1: Open the Project Folder

Open this folder:

```
C:\Coding\Pickleball_Lesson_Handouts
```

If using a terminal:

```
cd C:\Coding\Pickleball_Lesson_Handouts
```

### Step 2: Create a New Markdown Handout

Copy the starter template:

templates/new-handout-starter.md

Save the copy in the `examples` folder with a clear filename.

Example:

examples/serving-deep-student-handout.md

Use lowercase words separated by hyphens when naming files.

### Step 3: Generate the Handout Content

Use the prompt in:

prompts/create-student-handout.md

The simplest way is to give an AI assistant this instruction:

```
Create a student-facing pickleball lesson handout for: [LESSON TOPIC].
```

```
Use the template at:
```

```
C:\Coding\Pickleball_Lesson_Handouts\templates\student-handout-template.md
```

```
Follow these rules:
```

- Use the exact numbered section headings and emojis from the template.
- Write for beginner to intermediate adult students.
- Keep the tone clear, practical, encouraging, and coach-like.
- Put all practice drills and assignments in 7. Homework 🏠.
- Put quick pre-play reminders in 10. Quick Reminders 🧑.
- Put only outside references in 11. Resources 🔗.
- Do not duplicate drills or review lists in Resources.
- Use current web research for Resources.
- Include exactly three article links from three different sources.
- Include exactly three YouTube links from three different creators/channels.
- Include useful rule or strategy references only when they support the lesson.
- A numbered section must never split across pages. The build keeps each section together automatically; if a section is too tall to fit on one page, shorten it so it fits.
- Always visually inspect the rendered preview PNGs after building, and confirm no section is split across pages. Never rely on page count alone.
- Save the finished Markdown handout in:  
C:\Coding\Pickleball\_Lesson\_Handouts\examples\[lesson-slug]-student-handout.md

```
After writing the Markdown, build the PDF using:
```

```
python C:\Coding\Pickleball_Lesson_Handouts\scripts\build-handout-pdf.py  
C:\Coding\Pickleball_Lesson_Handouts\examples\[lesson-slug]-student-handout.md -o  
C:\Coding\Pickleball_Lesson_Handouts\outputs\[lesson-slug]-student-handout.pdf
```

```
Before delivering the PDF, visually inspect the generated preview PNGs in:
```

```
C:\Coding\Pickleball_Lesson_Handouts\work\[lesson-slug]-student-handout-preview
```

```
Confirm that major sections do not break awkwardly across pages and that Resources is present and coherent.
```

Replace `[LESSON TOPIC]` with the actual lesson topic.

Replace `[lesson-slug]` with a short lowercase filename version of the topic.

Example:

- Lesson topic: `Deep Serve and Return`
- Lesson slug: `deep-serve-and-return`

- Markdown file: `examples/deep-serve-and-return-student-handout.md`
- PDF file: `outputs/deep-serve-and-return-student-handout.pdf`

## Step 4: Build the PDF

From the project folder, run:

```
python .\scripts\build-handout-pdf.py .\examples\[lesson-slug]-student-handout.md -o .\outputs\[lesson-slug]-student-handout.pdf
```

Example:

```
python .\scripts\build-handout-pdf.py .\examples\serving-deep-student-handout.md -o .\outputs\serving-deep-student-handout.pdf
```

By default, no section is forced onto a new page. Each section is kept together and sections flow to fill each page, moving to the next page only when they will not fit as a whole.

## Step 5: Visually Check the PDF

After the script runs, check the generated preview images in:

`work\[lesson-slug]-student-handout-preview`

Open each `page-#.png` image and check:

- Section headings are not stranded at the bottom of a page.
- No numbered section is split across pages. Each section sits wholly on one page.
- Pages are reasonably filled; no large white-space gap is left where more sections could have fit.
- Resources is present and complete.
- Text is not clipped, overlapped, or too small.
- Link titles display correctly.
- The document feels polished and readable.

Do not rely only on page count or text extraction. Always visually inspect the preview images.

## Step 6: Review the Content

Use:

`docs/production-checklist.md`

Confirm:

- All 11 sections are present.
- The lesson topic is clear.
- Homework includes the drills.
- Quick Reminders includes only short pre-play cues.
- Resources includes only outside references.
- Resources has exactly three articles and exactly three YouTube videos.
- Article sources are not repeated.

- YouTube creators/channels are not repeated.
- Rule or strategy references are useful and relevant.

## Step 7: Share the PDF

Use the finished PDF from:

```
outputs\
```

Example:

```
outputs\serving-deep-student-handout.pdf
```

## Common Commands

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Build the existing third-shot drop example:

```
python .\scripts\build-handout-pdf.py .\examples\third-shot-drop-student-handout.md -o .\outputs\third-shot-drop-student-handout.pdf
```

Build a new handout:

```
python .\scripts\build-handout-pdf.py .\examples\[lesson-slug]-student-handout.md -o .\outputs\[lesson-slug]-student-handout.pdf
```

Force Resources to start on a new page:

```
python .\scripts\build-handout-pdf.py .\examples\[lesson-slug]-student-handout.md -o .\outputs\[lesson-slug]-student-handout.pdf --page-start-sections 11
```

Force sections 7, 8, and 11 to start on new pages:

```
python .\scripts\build-handout-pdf.py .\examples\[lesson-slug]-student-handout.md -o .\outputs\[lesson-slug]-student-handout.pdf --page-start-sections 7,8,11
```

## Setup Requirements

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This project needs:

- Python 3.10 or newer
- Google Chrome or Microsoft Edge
- Python packages listed in `requirements.txt`

Install the Python packages from inside the project folder:

```
python -m pip install -r requirements.txt
```

If the script cannot create preview images, install the requirements again:

```
python -m pip install pypdf pymupdf
```

## Caveats and Special Notes

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- Resources must be researched each time because links, article availability, YouTube titles, and view counts can change.
- The AI assistant should browse the web when creating the Resources section.
- The Resources section should not include practice drills or repeated review lists.
- Do not edit the PDF directly.
- Do not manually change fonts, colors, or spacing inside individual handouts.
- If the visual style needs to change, edit `templates/handout-style.css`.
- If the handout structure needs to change, edit `templates/student-handout-template.md`.
- If page breaks look awkward, rebuild with a different `--page-start-sections` value.
- Always inspect the rendered PNG previews before sending the PDF to students.
- The `work` folder is generated output and is intentionally ignored by Git.

## Publishing Changes to GitHub

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After adding or updating handouts, commit and push the changes:

```
git status
git add .
git commit -m "Add [lesson topic] handout"
git push
```

Use a clear commit message, such as:

```
git commit -m "Add deep serve handout"
```

## Quick Summary

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1. Copy `templates/new-handout-starter.md`.
2. Save it in `examples/` with a lesson-specific filename.
3. Use `prompts/create-student-handout.md` to generate the content.
4. Build the PDF with `scripts/build-handout-pdf.py`.
5. Check the preview PNGs in `work/`.
6. Review with `docs/production-checklist.md`.
7. Share the PDF from `outputs/`.